spiral arm A spiral-shaped structure extending outward from the central bulge of a spiral or barred spiral galaxy. It consists of gas, dust, emission nebulae, and hot young stars,

spiral galaxy A galaxy that consists of a spheroidal central concentration of stars (the **nuclear bulge**) surrounded by a flattened disk composed of stars, gas, and dust, within which the major visible features are clumped together into a pattern of spiral arms. See also *galaxy, spiral arm*.

star A self-luminous body of hot plasma that generates energy by means of nuclear fusion reactions.

starburst galaxy A galaxy within which star formation is taking place at an exceptionally rapid rate.

star cluster A group of between a few tens and around 1 million stars held together by gravity. All the member stars of a particular cluster are thought to have formed from the same original massive cloud of gas and dust. There are two principal types of cluster: open clusters and globular clusters. See also *globular cluster*, *open cluster*.

stellar-mass black hole see *black hole*.

stellar parallax see *parallax*.

stellar wind An outflow of charged particles from the atmosphere of a star. See also *solar wind*.

sun dog One of a pair of colored patches of light that sometimes may be seen on either side of the Sun, separated from the Sun by an angle of about 22°. Otherwise known as a **parhelion** or **mock sun**, a sun dog is formed when ice crystals in Earth's atmosphere refract sunlight. A **moon dog**, or **parselene**, is a patch of light that sometimes forms by the same process on either side of the Moon. A **parhelic circle** is a large, faint ring of white light, produced by the reflection of sunlight from atmospheric ice crystals, which crosses the Sun, passes through a pair of sundogs, and extends around the sky. Although a complete circle may be seen occasionally, more usually it is only possible to see arcs of light extending outward from the sundogs.

sunspot A patch on the surface of the Sun that appears dark because it is cooler than its surroundings. Sunspots occur in regions where localized concentrated magnetic fields impede the outward flow of energy from the solar interior. See also *solar cycle*.

supergiant An exceptionally luminous star with a very large diameter. Supergiant stars appear at the top of the Hertzsprung–Russell diagram. See also *Hertzsprung–Russell diagram*.

superior conjunction see *conjunction*.

superior planet A planet that travels around the Sun on an orbit that is outside the orbit of Earth. The superior planets are Mars, Jupiter, Saturn, Uranus, Neptune, and Pluto. See also *inferior planet*.

supermassive black hole see *black hole*.

supernova (plural: supernovae) A catastrophic event that destroys a star and causes its brightness to increase, temporarily, by a factor of around 1 million. A **type II supernova** occurs when the core of a massive star collapses and the rest of the star's material is blasted away; the collapsed core usually becomes a neutron star. A **type Ia supernova** involves the complete destruction of a white dwarf. The expanding cloud of debris from a supernova is called a **supernova remnant**. See also *neutron star*, *white dwarf*.

synchrotron radiation Electromagnetic radiation that is emitted when electrically charged particles (usually electrons) gyrate at very high speed around lines of force in a magnetic field. Synchrotron radiation has a characteristic continuous spectrum that is different from that which is emitted by a star or a black body. Astronomical sources of synchrotron radiation include supernova remnants and radio galaxies. See also *black body*, *electromagnetic radiation*, *spectrum*.

synchronous rotation The rotation of a body around its axis in the same period of time that it takes to orbit another body. Synchronous rotation, which is also known as **captured rotation**, is caused by tidal forces acting between the two bodies. Because its rotational and orbital periods are the same, the orbiting body always keeps the same face turned toward the object around which it is revolving. Like most of the planetary satellites, Earth's moon displays synchronous rotation. See also *orbital period*, *satellite*.

T

tail (of a comet) A stream, or streams, of ionized gas and dust that is swept out of the head of a comet (the coma) when it approaches, and begins to recede from, the Sun. A **type I tail** (or **gas tail**) consists of ionized gas driven out of the coma by the solar wind. A **type II tail** (or **dust tail**) is composed of dust particles that have been swept out of the coma by the pressure of sunlight. See also *comet*.

tectonic plate One of the large, rigid sections into which Earth's lithosphere (which comprises the crust and the rigid uppermost layer of Earth's mantle) is divided. Carried along by slow convection currents in the mantle, tectonic plates drift very slowly across the surface of the planet. Their relative motions give rise to phenomena such as earthquakes, volcanic activity, and mountain-building. The term "tectonic" is sometimes also used to refer to large-scale geological structures, and features resulting from their movement, on planets other than Earth. See also *convection*, *crust*, *mantle*.

tektite A small, rounded, glassy object formed when a large meteorite or asteroid strikes a rocky planet, melting the surface rocks and throwing molten drops of rock into the atmosphere. Typically a few inches (centimeters) across, tektites have been shaped by their flight through the atmosphere. On Earth's surface, they are found in a number of specific locations, called **strewn fields**. See also *asteroid*, *meteorite*.

terrestrial planet see *rocky planet*.

transit **(1)** The passage of a particular celestial body across an observer's meridian. **(2)** The passage of a body in front of a larger one (for example, the passage of the planet Venus across the face of the Sun, or a satellite across the face of a planet).

T Tauri star A young star, surrounded by gas and dust, that varies in brightness and usually shows evidence of a strong stellar wind (a stream of gas flowing away from the star). T Tauri stars are believed still to be contracting toward the main sequence. They are named after the first star of this kind to be identified. See also *main sequence*, *protostar*.

UV

ultraviolet radiation Electromagnetic radiation with wavelengths shorter than visible light but longer than X-rays. The hottest stars radiate strongly at ultraviolet wavelengths.

umbra **(1)** The dark, central cone of the shadow cast by an opaque body. The illuminating source will be completely hidden from view at any point within the umbra. **(2)** The darker, cooler central region of a sunspot, where the temperature is about 2,700–3,600°F (about 1,500–2,000°C) cooler than the average for the solar surface. See also *eclipse*, *penumbra, sunspot*.

vacuum energy see *cosmological constant*.

Van Allen belts Two concentric doughnut-shaped zones that contain charged particles (electrons and protons) trapped in Earth's magnetic field. They were discovered in 1958 by American space scientist James Van Allen.

variable star A star that varies in brightness. A **pulsating variable** is a star that expands and contracts in a periodic way, varying in brightness as it does so. An **eruptive variable** is a star that brightens and fades abruptly. A **cataclysmic variable** is a star that suffers one or more major explosions (for example, a nova). See also *Cepheid variable*, *nova*.

vernal equinox see *equinox*.

volcanic crater see *crater*.

W

wavelength The distance between two successive crests or between two successive troughs in a wave motion.

WIMP The acronym for **Weakly Interacting Massive Particle**, one of a range of postulated elementary particles that have high masses (tens or hundreds of times as great as that of a proton) but interact so exceedingly weakly with ordinary matter that they have not yet been directly detected. WIMPs are widely considered to comprise the major part of the dark-matter content of the universe. See also *dark matter*.

white dwarf star A star of low luminosity but relatively high surface temperature that has ceased to generate energy by nuclear-fusion reactions, that has been compressed by gravity to a diameter comparable to that of Earth, and that is slowly cooling and fading. See also *black dwarf*, *Hertzsprung–Russell diagram*.

Wolf–Rayet star A very hot star from which gas is escaping at an exceptionally rapid rate, which is surrounded by an expanding gaseous envelope, and which has emission lines in its spectrum. See also *emission line*, *spectrum*.

XYZ

X-ray burster An object that emits strong bursts of X-rays, lasting from a few seconds to a few minutes. The bursts are believed to occur when gas drawn from an orbiting companion star accumulates on the surface of a neutron star and triggers a nuclear-fusion chain reaction. See also *fusion*, *neutron star*.

X-ray radiation Electromagnetic radiation with wavelengths shorter than ultraviolet radiation but longer than gamma rays. X-rays are emitted by extremely hot clouds of gas, such as the solar corona.

zenith The point on the sky directly above an observer (that is, 90° above the observer's horizon).

zodiac A band around the celestial sphere that extends for 9° on either side of the ecliptic, and through which the Sun, Moon, and naked-eye planets appear to travel. The zodiac contains part or all of 24 constellations. In the course of the year, the Sun passes through 13 of these constellations, 12 of which correspond to the astrological "signs of the zodiac." See also *ecliptic*.

zodiacal light A faint, cone-shaped glow that extends along the direction of the ecliptic from the western horizon after sunset or from the eastern horizon before sunrise. Most easily seen from tropical skies, it is caused by the scattering of sunlight by particles of interplanetary dust that lie close to the plane of the ecliptic.